

Boss Challenge — Paper 54 (Class 7)

Reproduction in Plants | Winds, Storms & Cyclones | Algebraic Expressions | Fractions & Decimals
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A potato left in a basket sprouts new shoots from the buds (the 'eyes') on its surface. This way of making new plants without seeds is called:
(A) vegetative propagation
(B) pollination
(C) germination of a seed
(D) fertilisation
2. A balloon stays firm and rounded when blown up because the air trapped inside pushes outward on its walls. This shows that air:
(A) exerts pressure
(B) cannot move
(C) is solid
(D) has no weight
3. In algebra a letter such as x , which can stand for different numbers at different times, is called a:
(A) variable
(B) equation
(C) coefficient
(D) constant
4. Adding the two fractions one-half and one-quarter together, the sum $\frac{1}{2} + \frac{1}{4}$ comes to:
(A) $\frac{1}{2}$
(B) $\frac{2}{6}$
(C) $\frac{3}{4}$
(D) $\frac{1}{4}$
5. The part of a flower that makes the dust-like pollen grains is the anther, which is held up on a stalk. The anther together with its stalk is the:
(A) sepal
(B) petal
(C) stamen
(D) pistil
6. Whenever a portion of air is heated, it expands, becomes lighter than the cooler air around it, and therefore tends to:
(A) stay exactly in place
(B) turn into water
(C) sink to the floor
(D) rise upward

7. In the algebraic term $5x$, the number 5 that multiplies the variable x is known as its:
- (A) variable
 - (B) exponent
 - (C) constant
 - (D) coefficient
8. Taking one-half of one-third, the product $\frac{1}{2} \times \frac{1}{3}$ works out to:
- (A) $\frac{3}{2}$
 - (B) $\frac{1}{6}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{2}{5}$
9. When pollen grains move from a flower's anther across to the stigma, whether on the same bloom or another, this process is called:
- (A) pollination
 - (B) fertilisation
 - (C) respiration
 - (D) germination
10. Wind always blows from a place where the air pressure is higher towards a place where the air pressure is:
- (A) frozen
 - (B) lower
 - (C) exactly equal
 - (D) higher
11. Collecting the like terms $4x$ and $3x$ into a single term, the sum $4x + 3x$ equals:
- (A) 7
 - (B) $7x^2$
 - (C) $7x$
 - (D) $12x$
12. Adding the two decimals 0.5 and 0.25, the sum $0.5 + 0.25$ comes to:
- (A) 0.75
 - (B) 0.30
 - (C) 0.7
 - (D) 7.5
13. A pod splits open to reveal 8 seeds, and exactly 6 of them are healthy enough to sprout. The fraction of the seeds that will sprout is:
- (A) $\frac{6}{8}$
 - (B) $\frac{1}{4}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{8}{6}$

14. Holding a paper strip to your lips and blowing hard over its upper surface makes the free end lift up. This happens because fast-moving air has a:
- (A) no effect on pressure
 - (B) higher pressure
 - (C) lower pressure
 - (D) a solid surface
15. An algebraic expression made up of just a single term, such as $7y$, is given the special name:
- (A) trinomial
 - (B) equation
 - (C) binomial
 - (D) monomial
16. Finding one-half of 6 by multiplying, the value of $\frac{1}{2} \times 6$ is:
- (A) 6
 - (B) 3
 - (C) 12
 - (D) 2
17. In the tiny fungus called yeast, a small bulge grows out of the parent cell and then pinches off to live on its own. This form of asexual reproduction is:
- (A) budding
 - (B) fertilisation
 - (C) pollination
 - (D) fragmentation
18. A cyclone is a huge weather system built around a centre of very low air pressure, with strong winds spiralling inwards. The calm region at its very centre is called the:
- (A) eye
 - (B) crest
 - (C) tail
 - (D) front
19. Counting the separate terms in the expression $3x + 4y - 7$, the number of terms is:
- (A) two
 - (B) three
 - (C) one
 - (D) seven
20. Adding the decimals 1.5 and 2.5 together, the sum $1.5 + 2.5$ equals:
- (A) 3.5
 - (B) 4
 - (C) 3.10
 - (D) 4.10

21. A pond plant called Spirogyra grows into a long thread that breaks into smaller pieces, and each piece grows into a new plant. This kind of asexual reproduction is:

- (A) germination
- (B) pollination
- (C) fragmentation
- (D) budding

22. Light, dry seeds fitted with wings or tufts of hair, such as those of the drumstick or madar plant, are carried far from the parent mainly by:

- (A) water
- (B) sound
- (C) wind
- (D) animals

23. Writing 'five more than a number n ' as an algebraic expression gives:

- (A) $5n$
- (B) $5 - n$
- (C) $n + 5$
- (D) $n - 5$

24. A gardener sows 200 seeds and 0.6 of them come up as seedlings. The number of seeds that germinated is:

- (A) 60
- (B) 200
- (C) 6
- (D) 120

25. After fertilisation inside a flower, the ovule grows and develops into the:

- (A) fruit
- (B) seed
- (C) flower
- (D) root

26. A wide stretch of warm ocean adds large amounts of water vapour and heat to the air above it, which is why violent cyclones most often build up over:

- (A) dry deserts
- (B) warm seas near the coast
- (C) deep underground caves
- (D) cold mountain tops

27. Replacing x with 3 in the expression $x + 5$, its value becomes:

- (A) 2
- (B) 35
- (C) 8
- (D) 15

28. The female reproductive part of a flower, made up of the stigma, the style and the ovary, is called the:

- (A) petal
- (B) stamen
- (C) pistil
- (D) sepal

29. An open tin can with a little water is boiled, then sealed and cooled. It suddenly crumples inward. The can is crushed because, after the steam cools, it is pushed by the:

- (A) the metal melting
- (B) a magnet pulling it
- (C) greater air pressure outside the can
- (D) water boiling again inside

30. Substituting $x = 4$ into the expression $2x + 3$, the value of the expression is:

- (A) 14
- (B) 9
- (C) 24
- (D) 11

31. The reciprocal of the fraction $\frac{2}{3}$ (the fraction you turn it upside down to get) is:

- (A) $\frac{1}{6}$
- (B) $\frac{3}{2}$
- (C) 6
- (D) $\frac{2}{3}$

32. Flowers that are brightly coloured, sweetly scented and rich in nectar are usually adapted to be pollinated by:

- (A) insects
- (B) water
- (C) wind
- (D) sunlight

33. At a weather station, the device with spinning cups that records how fast the wind blows is the:

- (A) anemometer
- (B) barometer
- (C) rain gauge
- (D) thermometer

34. Naming the type of expression that has exactly two terms, such as $x + 7$, we call it a:

- (A) trinomial
- (B) monomial
- (C) variable
- (D) binomial

35. Dividing one-half by one-quarter, the value of $1/2 \div 1/4$ is:

- (A) 4
- (B) $1/8$
- (C) $1/2$
- (D) 2

36. After fertilisation, while the ovule becomes the seed, the ovary of the flower grows and ripens into the:

- (A) seed
- (B) fruit
- (C) petal
- (D) root

37. During a thunderstorm, warm moist air rises rapidly, cools high up, and its water vapour turns into water droplets. This change of vapour into droplets is called:

- (A) germination
- (B) filtration
- (C) evaporation
- (D) condensation

38. The number of seeds in one pod is the same in every pod of a plant. If each pod has p seeds, the total number of seeds in 6 such pods is:

- (A) 6
- (B) $p + 6$
- (C) $p/6$
- (D) $6p$

39. Comparing the two fractions $1/2$ and $1/3$, the larger fraction is:

- (A) $1/3$
- (B) $1/2$
- (C) $1/6$
- (D) they are equal

40. Some seeds are covered with tiny hooks or stiff spines that catch onto the fur of passing animals and are carried away. These seeds are dispersed by:

- (A) sunlight
- (B) animals
- (C) wind
- (D) water

41. A steady fall in the reading of a barometer (the air pressure dropping over a few hours) is often a useful warning that there may soon be:

- (A) a rainbow
- (B) a clear sunny day
- (C) a cold frost
- (D) a storm

42. Identifying the constant term in the expression $3x + 7$, the term that has no variable is:

- (A) 3
- (B) $3x$
- (C) 7
- (D) x

43. Writing the decimal 0.1 as a fraction, it equals:

- (A) $\frac{1}{10}$
- (B) $\frac{1}{100}$
- (C) 10
- (D) 1

44. A flower that contains BOTH stamens (male part) and a pistil (female part) within the same flower is described as a:

- (A) bisexual flower
- (B) unisexual flower
- (C) petal-less flower
- (D) seedless flower

45. Coastal towns hit by cyclones often suffer flooding and great damage because a cyclone brings together very high-speed winds and:

- (A) a fall in temperature
- (B) very heavy rainfall
- (C) bright sunshine
- (D) dry dusty air

46. A young plant grows taller by 2 cm every day. The total growth, in centimetres, after d days is given by the expression:

- (A) $\frac{d}{2}$
- (B) $d + 2$
- (C) $2 + d$
- (D) $2d$

47. Working out three-quarters of 20, the value of $\frac{3}{4} \times 20$ is:

- (A) 15
- (B) 60
- (C) 23
- (D) 5

48. Cross-pollination takes place when pollen from one flower is carried to the stigma of a flower on a different plant of the same kind. Self-pollination, by contrast, happens when pollen reaches the stigma of:

- (A) the same flower (or another flower on the same plant)
- (B) a flower on a completely different plant
- (C) the soil around the roots
- (D) a flower of a different species

49. High windows or ventilators are placed near the ceiling of a room mainly so that the warm air, which collects at the top because it is lighter, can:

- (A) trap the warm air inside
- (B) stop any air from moving
- (C) keep cold air from rising
- (D) escape and let the room stay cool

50. Forming the expression for 'three less than twice a number x ', we write:

- (A) $2x - 3$
- (B) $3 - 2x$
- (C) $2x + 3$
- (D) $2(x - 3)$

51. Over two days a rain gauge collected 2.5 cm of rain on the first day and 1.5 cm on the second. The total rainfall for the two days was:

- (A) 3.10 cm
- (B) 40 cm
- (C) 4 cm
- (D) 1 cm

52. Tiny new plantlets sprout along the notched edges of the thick leaf of a Bryophyllum plant, drop off and grow into new plants. This is an example of reproduction by:

- (A) fertilisation
- (B) pollination
- (C) vegetative propagation through leaves
- (D) seed germination

53. Seeds and fruits that are light and can float, like the coconut with its thick fibrous husk, are most often carried to new places by:

- (A) water
- (B) magnets
- (C) animals
- (D) wind

54. Combining the like terms in the expression $3a + 2b + 5a$, the simplified form is:

- (A) $8a + 2b$
- (B) $5ab + 5a$
- (C) $10ab$
- (D) $8a + 5a$

55. Writing the fraction $5/10$ in its simplest form, it reduces to:

- (A) $1/5$
- (B) $1/2$
- (C) $5/10$
- (D) $2/1$

56. The sticky tip at the top of the pistil, which receives the pollen grains during pollination, is the:
- (A) anther
 - (B) stigma
 - (C) filament
 - (D) ovary
57. On a hot afternoon at the seaside, the land heats up faster than the sea, so cool air moves in from the sea towards the land. This daytime wind is called the:
- (A) cyclone
 - (B) land breeze
 - (C) monsoon wind
 - (D) sea breeze
58. A model for a gust of wind gives its speed in km/h as $20 + 5t$, where t is the time in hours. When $t = 4$, the wind speed is:
- (A) 100 km/h
 - (B) 29 km/h
 - (C) 25 km/h
 - (D) 40 km/h
59. Multiplying the two decimals 0.2 and 0.3, the product 0.2×0.3 equals:
- (A) 0.06
 - (B) 6
 - (C) 0.6
 - (D) 0.5
60. Seeds are usually scattered far away from the parent plant. The main advantage of this dispersal is that the new seedlings:
- (A) stay close together for warmth
 - (B) grow faster than every other plant
 - (C) do not crowd and compete with the parent for water, light and space
 - (D) never need any sunlight
61. The fusion of the male gamete (from a pollen grain) with the egg inside the ovule, which happens after pollination, is called:
- (A) transpiration
 - (B) fertilisation
 - (C) pollination
 - (D) germination
62. Out of the terms in the expression $2x - 5$, the two separate terms are:
- (A) $2x$ and 5
 - (B) $2x$ and -5
 - (C) x and -5
 - (D) 2 and x

63. Among the three decimals 0.5, 0.05 and 0.55, the greatest in value is:

- (A) 0.5
- (B) 0.55
- (C) they are all equal
- (D) 0.05

64. A flower that carries only stamens, or only a pistil, but not both kinds of reproductive part, is described as a:

- (A) bisexual flower
- (B) unisexual flower
- (C) scented flower
- (D) seedless flower

65. Air that moves from one place to another, which we can feel on our skin and which makes flags flutter, is simply:

- (A) vapour
- (B) wind
- (C) sound
- (D) pressure

66. Writing 'twice a number x' as an algebraic expression gives:

- (A) $2x$
- (B) x^2
- (C) $x + 2$
- (D) $x/2$

67. Writing the fraction $7/10$ as a decimal number, it equals:

- (A) 0.7
- (B) 70
- (C) 7.0
- (D) 0.07

68. A bread mould spreads to fresh bread by releasing huge numbers of tiny, dust-like reproductive units into the air. These units are called:

- (A) roots
- (B) spores
- (C) buds
- (D) seeds

69. On a clear night at the coast the land cools faster than the sea, so air now blows from the land out towards the sea. This night-time wind is the:

- (A) cyclone
- (B) land breeze
- (C) monsoon
- (D) sea breeze

70. Writing 'a number x decreased by 4' as an algebraic expression gives:

- (A) $x - 4$
- (B) $x + 4$
- (C) $4x$
- (D) $4 - x$

71. Writing the decimal 0.75 as a fraction in its simplest form, it equals:

- (A) $\frac{7}{5}$
- (B) $\frac{3}{4}$
- (C) $\frac{1}{4}$
- (D) $\frac{75}{10}$

72. Once a pollen grain settles on the stigma, it grows a fine tube down through the style. The job of this pollen tube is to carry the male gamete to the:

- (A) petal
- (B) root
- (C) ovule (containing the egg)
- (D) sepal

73. Because the Sun heats the Earth unevenly, the air over some regions becomes warmer than over others. This uneven heating of the air is the basic cause of:

- (A) earthquakes
- (B) eclipses
- (C) tides
- (D) winds

74. Writing 'the product of 3 and y ' as an algebraic expression gives:

- (A) $3y$
- (B) $y - 3$
- (C) $3 + y$
- (D) $y/3$

75. Working out one-fifth of 100, the value of $\frac{1}{5} \times 100$ is:

- (A) 50
- (B) 100
- (C) 5
- (D) 20

76. Flowers that depend on the wind to carry their pollen are usually small, dull-coloured and without scent or nectar. They also tend to make:

- (A) brightly coloured petals
- (B) large amounts of light pollen
- (C) sweet nectar to attract bees
- (D) very few heavy pollen grains

77. When a cyclone warning is given for a coastal area, the safest and most sensible action for people there is to:

- (A) ignore the warning and carry on
- (B) go to the beach to watch the waves
- (C) open all the windows wide
- (D) move to a safe shelter and follow official advice

78. Subtracting the like term $2x$ from $5x$, the difference $5x - 2x$ equals:

- (A) $3x$
- (B) 3
- (C) $7x$
- (D) $10x$

79. Adding the fractions one-third and one-third, the sum $\frac{1}{3} + \frac{1}{3}$ equals:

- (A) $\frac{1}{3}$
- (B) $\frac{1}{6}$
- (C) $\frac{2}{3}$
- (D) $\frac{2}{6}$

80. Seeds are produced only after a male gamete fuses with a female egg cell. Because it needs this fusion of two gametes, seed formation is an example of:

- (A) vegetative propagation
- (B) budding
- (C) sexual reproduction
- (D) asexual reproduction

81. A violently spinning column of air with a narrow funnel shape, which forms over land and can suck up dust and objects, is called a:

- (A) dewfall
- (B) rainbow
- (C) tornado
- (D) sea breeze

82. Replacing x with 3 in the expression $10 - 2x$, the value of the expression is:

- (A) 6
- (B) 16
- (C) 4
- (D) 8

83. Multiplying the decimal 2.5 by 2, the product 2.5×2 equals:

- (A) 25
- (B) 2.10
- (C) 4.10
- (D) 5

84. In a brightly coloured flower, the showy parts whose main job is to attract insects for pollination are the:

- (A) sepals
- (B) anthers
- (C) petals
- (D) ovules

85. An 'empty' bottle is pushed mouth-down into a tub of water and bubbles escape from it. This shows that the bottle was not really empty but was filled with air, proving that air:

- (A) has a colour
- (B) takes up space (occupies volume)
- (C) is a liquid
- (D) cannot move

86. In the algebraic term $7m$, the part that can take different values - the variable - is:

- (A) m
- (B) $7m$
- (C) $+$
- (D) 7

87. Finding one-half of 0.5, the value of $\frac{1}{2} \times 0.5$ is:

- (A) 1
- (B) 0.5
- (C) 0.025
- (D) 0.25

88. Before a flower opens, its delicate inner parts are wrapped and protected by the small, usually green, leaf-like structures called the:

- (A) stamens
- (B) stigma
- (C) sepals
- (D) petals

89. During a thunderstorm we first see a flash of lightning and a little later hear the thunder. The flash comes first because, compared with sound, light travels:

- (A) at the same speed
- (B) much more slowly
- (C) only as fast as the wind
- (D) very much faster

90. An algebraic expression that has exactly three terms, such as $2x + 3y - 5$, is called a:

- (A) monomial
- (B) trinomial
- (C) variable
- (D) binomial

91. A cluster on a plant holds 12 buds, and exactly one-third of them have opened into flowers. The number of buds that have opened is:
- (A) 12
 - (B) 8
 - (C) 4
 - (D) 3
92. As a cyclone strengthened, its strongest wind speed rose from 90 km/h to 120 km/h. The increase in wind speed was:
- (A) 120 km/h
 - (B) 210 km/h
 - (C) 90 km/h
 - (D) 30 km/h
93. A pod holds s seeds. A botanist has 4 such pods plus 2 loose seeds. The total number of seeds, as an algebraic expression, is:
- (A) $4s$
 - (B) $6s$
 - (C) $4s + 2$
 - (D) $4 + 2s$
94. A plant bears 100 seeds, and one-quarter of them actually germinate into seedlings. The number of seeds that germinated is:
- (A) 100
 - (B) 4
 - (C) 25
 - (D) 50
95. Out of 25 seeds sown, three-fifths of them sprout. The number of seeds that sprouted is:
- (A) 15
 - (B) 25
 - (C) 5
 - (D) 10
96. Replacing x with 2 in the expression $3x - 1$, the value of the expression is:
- (A) 4
 - (B) 6
 - (C) 8
 - (D) 5
97. Adding the three equal fractions one-quarter, one-quarter and one-quarter, the sum $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$ equals:
- (A) $\frac{3}{12}$
 - (B) $\frac{1}{12}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{3}{4}$

98. A cyclone's centre moved a distance of 90 km across the sea in 3 hours. The speed at which the cyclone was moving was:

- (A) 3 km/h
- (B) 270 km/h
- (C) 90 km/h
- (D) 30 km/h

99. Plants raised from cuttings or tubers, rather than from seeds, flower earlier and are exactly like the parent plant. This sameness happens because such plants are produced by:

- (A) cross-pollination
- (B) pollination by insects
- (C) sexual reproduction
- (D) asexual reproduction (vegetative propagation)

100. Writing the fraction $\frac{4}{5}$ as a decimal number, it equals:

- (A) 0.8
- (B) 0.45
- (C) 5.4
- (D) 0.4